Lab 11 — Explore Microsoft Entra ID: Users and Groups

In this lab you explore your tenant in Microsoft Entra ID (formerly Azure AD), create a user and a security group, and sign in as that user to see what "authenticated but not authorised" looks like. This maps to the AZ-900 skill area **Describe Azure architecture and services** — identity, access, and the difference between authentication and authorization.

Run on https://portal.azure.com — sign in with your Azure free account (https://azure.microsoft.com/free).

Step 1 — WARNING: use your own tenant, not your company's

Do this lab only in the tenant of your own Azure free account. If your portal session is signed into a company or school tenant, creating users and groups there may violate policy and confuse your IT department.

1. Click your account avatar (top-right) > **Switch directory**.
2. Confirm the current directory is your personal free-account tenant (usually named "Default Directory" with a domain like `<yourname>outlook.onmicrosoft.com`). Switch to it if not.

Step 2 — Open Microsoft Entra ID and note your tenant details

1. In the portal search bar, type **Microsoft Entra ID** and open it.
2. On the **Overview** blade, write down:

- **Tenant name** (e.g. `Default Directory`)
- **Primary domain** — `<something>.onmicrosoft.com` (you need this exact value in Step 3)
- **Tenant ID** — a GUID

1. Note the counts of Users, Groups, and Applications.

Every Azure subscription trusts exactly one Entra tenant — the tenant is the identity boundary; the subscription is the billing/resource boundary.

Step 3 — Create a new user

1. In Entra ID, click **Manage > Users** > **+ New user** > **Create new user**.
2. On the **Basics** tab enter:

Field	Value
User principal name	`az900.learner` @ `<yourtenant>.onmicrosoft.com`

Field	Value
Display name	`AZ900 Learner`
Password	tick **Auto-generate password**, then click the copy icon and **save the password** — you need it in Step 6

1. Click **Review + create** > **Create**.
2. Refresh the Users list and confirm `AZ900 Learner` appears with source "Cloud".

This is a cloud-only identity that exists purely in your tenant — no on-premises AD involved.

Step 4 — Create a security group

1. In Entra ID, click **Manage > Groups** > **+ New group**.
2. Enter:

- Group type: `Security`
- Group name: `grp-az900-learners`
- Membership type: `Assigned`

1. Click **Create**.

Assigned membership means an admin adds members manually; **Dynamic** groups instead auto-populate from rules (e.g. department = Sales) — a Premium feature.

Step 5 — Add the user to the group

1. Open **grp-az900-learners** > **Manage > Members** > **+ Add members**.
2. Search for `AZ900 Learner`, select it, click **Select**.
3. Confirm the member count is now 1.

In real deployments you assign roles and licences to **groups**, not individual users — new hires inherit access just by joining the group.

Step 6 — Sign in as the new user

1. Open a **private/incognito** browser window and go to https://portal.azure.com.
2. Sign in as `az900.learner@<yourtenant>.onmicrosoft.com` with the auto-generated password.
3. You are forced to **update the password** — set a new one and note it (keep MFA enrolment prompts minimal if offered; you can select "Ask later" where available).

4. The portal opens. **Authentication succeeded** — the tenant verified who this user is.

Step 7 — Observe: authenticated but not authorised

Still in the private window as `az900.learner`:

1. Search **Subscriptions** — the list is empty ("You don't have any subscriptions").
2. Search **Resource groups** — no resource groups are visible, even though `rg-az900-labs` exists in the same tenant.
3. Try **Create a resource** > pick anything > the Basics tab has no subscription to select, so creation is impossible.

This is the key lesson: Entra ID **authenticated** the user (proved identity), but with no Azure **RBAC role assignments** the user is **authorised** to do nothing with resources. Lab 12 fixes this by assigning a role. Sign out of the private window but keep the user and its new password.

Step 8 — Look at authentication methods and MFA (look, don't enforce)

Back in your admin window:

1. In Entra ID, click **Manage > Security > Authentication methods** (or **Manage > Users** > select a user > **Authentication methods**).
2. Browse the available methods: Microsoft Authenticator, SMS, FIDO2 security keys, Temporary Access Pass.
3. **Do NOT enable any enforcement policy** — just note that multifactor authentication requires two or more of: something you know (password), something you have (phone/key), something you are (biometric).

Free-tier tenants also have **Security defaults** (Entra ID > Overview > Properties > Manage security defaults), which enforce MFA tenant-wide — leave it as is for this course.

Clean up

Keep the user `az900.learner` and the group `grp-az900-learners` — Lab 12 (RBAC) reuses them. They cost nothing.

If you are not continuing to Lab 12, delete them: Entra ID > **Users** > tick `AZ900 Learner` > **Delete**, and **Groups** > tick `grp-az900-learners` > **Delete**. Keep the resource group `rg-az900-labs` — it is reused until Lab 16.

What you learned

- **Microsoft Entra ID** is Azure's cloud identity service; every subscription trusts one tenant, identified by a primary domain and tenant ID.
- **Users** are identities; **security groups** (Assigned or Dynamic) are the unit you should manage access with.
- **Authentication** (proving who you are) is separate from **authorization** (what you may do) — a new user can sign in yet see zero subscriptions or resources.
- Azure resource access comes from **RBAC role assignments**, not from merely existing in the tenant (explored in Lab 12).
- **MFA** strengthens authentication by combining something you know, have, and are; Entra ID supports Authenticator, SMS, and FIDO2 keys.